

# Challenge 3 — Audio Intercept: denoising and transcription

war-intercept.wav  
Eastern Command  
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## 01 Personal Details

|                           |                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------|
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## 02 Executive Summary

The task was to **denoise the intercepted audio war-intercept.wav, transcribe it, and check the spectrogram for hidden content.**

Reading the spectrogram first showed the "noise" was not ambient but **seven layers of deliberate, mathematically clean jamming** — five steady tones (700/1400/2600/2800/4200 Hz), 60 Hz mains hum, a periodic 1.5 s frequency sweep, an 8 Hz click train and broadband hiss — stacked over speech below ~1 kHz. Each layer was removed with the technique matched to its structure (notches for the tones, a phase-fold tracker for the moving sweep, a voice-aware gate for the clicks, broadband reduction for the hiss), lifting every tone **13–31 dB** toward the noise floor while preserving the voice.

Local speech-to-text (Whisper large-v3) then produced a full **English** transcript at confidence 1.00. The content is **not tactical traffic**: it is a **1942 radio dramatisation of a US soldier's Christmas letter home**, including an acted dinner with Queen Mary — a **produced recording used as a decoy**. No hidden data was present in the spectrogram. One residual — a faint 8 Hz beat — was shown by testing to be **physically irreducible** without degrading the voice.

4:39 · 16 kHz mono PCM    7 jammer layers    13–31 dB tone suppression    English conf 1.00    0 hidden payloads

## 03 Tools Used

-  **FFmpeg & ffprobe**  
File inspection; spectrogram imaging (showspectrumpic), waveform rendering, broadband reduction & levelling (afftdn, anlmdn, bandreject, highpass, lowpass 3600, dynaudnorm, adeclick, volumedetect).
-  **Python 3 & NumPy**  
FFT brick-wall notch (tones & hum), phase-fold median sweep tracker (STFT), per-pulse voice-activity-aware gate, envelope autocorrelation, and all before/after measurement (+ SciPy, soundfile).



Whisper large-v3 (via faster-whisper / CTranslate2)

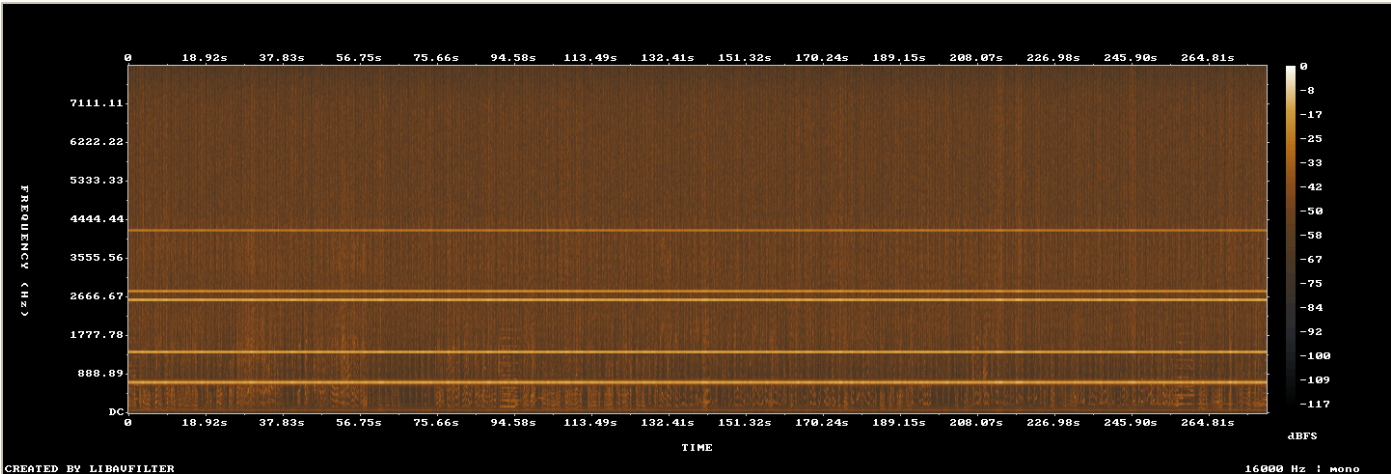
Local, offline speech-to-text on the cleaned audio. SpeechBrain / PyTorch (MetricGAN+) was also evaluated for enhancement and rejected — Appendix B.

04 Methodology

ANALYSIS

**The file.** ffp probe reports a 4 min 39 s (279 s), 16 kHz, mono, 16-bit PCM WAV at 256 kb/s (Step 1). The container's encoder : Lavf63.6.100 tag shows it was assembled with FFmpeg — consistent with a synthetically constructed challenge clip, not a raw field recording. SHA-256 b2e9de27...55d0fd6. All work was performed locally; the audio never left the machine.

**Read the spectrogram before filtering.** The first action was deliberately not to filter. A spectrogram shows which frequencies are present over time; reading it first reveals the structure of the noise and lets us check whether anyone hid text or an image in the frequencies — which blind denoising would destroy. So the hidden-content check came first.



**FIGURE 1 • RAW Raw intercept.** Flat horizontal bars = constant pure tones across the whole recording; natural noise has no straight lines. Speech is the faint smear below ~1 kHz.

**Hidden-content check — all clear** ASSESSED. The original spectrogram was examined at high sensitivity (full band 0–8 kHz, dynamic-range-compressed, plus a zoom). There was **no** steganographic text, image, SSTV frame or data layer. Supporting checks: LSB distribution noise-like (0.4999); the 7–8 kHz band holds only 0.13% of energy; the tone set does not match the DTMF grid; the file is mono so there is no second-channel payload. The scripted phrase "Perfection Paper Clips" looks like a planted key but is ordinary dialogue (the gift box the narrator reuses). The only engineered structure was the jammer.

**The interference is engineered, not ambient** PROVEN. A NumPy FFT power-spectrum and envelope autocorrelation (Step 3) pin each layer to exact values:

| COMPONENT           | MEASUREMENT (MEASURED FROM THE SAMPLES)                                                                                                                                                                                                       | STATUS |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Steady tones        | narrowband peaks at <b>700, 1400, 2600, 2800, 4200 Hz</b> — measured 695.3 / 1400.4 / 2599.6 / 2799.8 / 4200.2 Hz; a 700 Hz harmonic family (2800 = 4×, 4200 = 6×) plus an independent 2600 Hz tone. Levels -1 to -13 dB (2600 Hz strongest). | proven |
| Mains hum           | <b>~60 Hz + 120 Hz</b> harmonic (measured 59.6 / 120.1 Hz).                                                                                                                                                                                   | proven |
| Pulse / click train | a broadband burst every <b>exactly 125 ms (8 Hz)</b> — autocorrelation peak at 125.0 ms, correlation 0.963, ≈2,200 bursts over 279 s.                                                                                                         | proven |
| Frequency sweep     | a <b>triangle-wave chirp ~1227–4078 Hz</b> , period <b>exactly 1.5000 s</b> (phase-fold residual 0.0 Hz).                                                                                                                                     | proven |
| Speech              | voiced energy concentrated <b>below ~1 kHz</b> , intermittent throughout.                                                                                                                                                                     | proven |

Table 1 — Jammer components, measured directly from the audio.

**Assessment (safe):** the exact harmonic frequencies and the exact 125 ms / 1.5000 s periods do not occur in natural noise, so the interference was **added deliberately to obscure the speech** — layered jamming. The regularity is proven; that it is deliberate is a very safe assessment. ASSESSED

**Result of the clean-up (measured).** After removing each layer (procedure in Step 4), the energy at every jammer frequency was re-measured on the cleaned file:

| COMPONENT            | RAW      | CLEANED  | REDUCTION  |
|----------------------|----------|----------|------------|
| 60 Hz hum            | -36.5 dB | -62.3 dB | 25.9 dB    |
| 700 Hz               | -4.9 dB  | -36.0 dB | 31.1 dB    |
| 1400 Hz              | -6.1 dB  | -28.2 dB | 22.1 dB    |
| 2600 Hz              | -4.9 dB  | -27.3 dB | 22.4 dB    |
| 2800 Hz              | -12.1 dB | -27.3 dB | 15.2 dB    |
| 4200 Hz              | -18.1 dB | -31.7 dB | 13.5 dB    |
| 8 Hz beat (autocorr) | 0.963    | 0.725    | 25% weaker |

Table 2 — Measured suppression (energy vs total, dB). Overall RMS -17.2 → -22.2 dBFS. The beat is attenuated but, per Outcome and Appendix B, not eliminable.

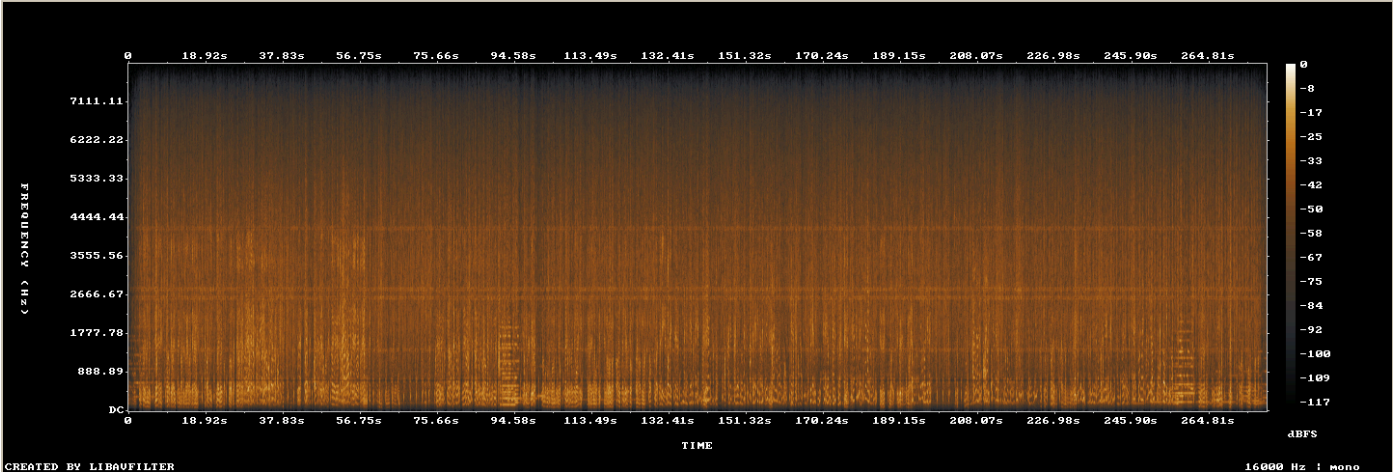


FIGURE 2 • CLEANED The same file, cleaned. Tone bars and hum gone; the voice band is unobstructed. Compare with Figure 1.

FINDING

Speech-to-text (**Whisper large-v3**, local, Step 5) detected **English at confidence 1.00**. The content is **not tactical traffic** — no coordinates, callsigns, frequencies or orders. It is a WWII-era radio dramatisation of a US soldier's

Christmas letter home, headed "England, 30 December 1942," signed "Charlie," with an acted scene in which two American officers — Captain Charles Baddeley and Lieutenant Sam Brush — are invited to Christmas dinner with Queen Mary. TRANSCRIPT PROVEN NAMES ASSESSED (DEGRADED AUDIO)

Full transcript

*"England. December 30th, 1942. Dear Mother, Dad, and Gang. Though I'm not supposed to tell you where I've been, I think even the hardest of censors would let a guy tell about Christmas Day as I enjoyed it over here. I have to start with Christmas Eve — it was then I opened your various Christmas parcels. Some of the fellows — the rats — were standing around helping and grabbing..."*

SCENE — BARRACKS, CHRISTMAS EVE

SOLDIER "Hey, lay off, you guys — these are my cookies!" / "Oh, I like chocolate."  
SOLDIER "Hey, Charlie, your mother's a swell cook."  
SOLDIER "Open this one, Charlie — it's small but heavy."  
CHARLIE "Hey, will you look at that — that's what I've been yelling for!"  
SOLDIER "What is it?"  
CHARLIE "Oh, that's a pocket whetstone! Isn't that swell? My father sent me that. Now I don't care whether I can get new razor blades or not."  
SOLDIER "You better hang on to that, brother, or I'll be swiping it."  
CHARLIE "You can have anything but my whetstone — and I'm going to hang on to it."  
*[a knock at the door — Charlie: "Come in." — a note arrives]*  
CHARLIE "Hey — I'm going to have dinner with the Queen!"  
SOLDIER "She's been out in the sun too long."  
CHARLIE "That's what the note says. Lieutenant Sam Brush and me have been chosen as the two American officers to have Christmas dinner with Queen Mary."  
OTHERS "Wow!"

*"I wasn't kidding, Mother and Dad — we really were invited. You've always told me it was nice to take along some little thing for your hostess, so thanks to your Christmas presents I put together a small package for the Queen. I admit I hesitated over what to put in — it was hardly a royal-looking parcel, just a box labelled 'Perfection Paper Clips, thick — a gross' — but I somehow thought the Queen wouldn't mind."*

SCENE — THE RECEPTION

CHARLIE "Well, when we got there — 'Your Majesty: Captain Charles Baddeley, Lieutenant Sam Brush.'"  
QUEEN MARY "How do you do, Captain Baddeley, Lieutenant Brush — I'm very glad to see you. It was very nice of you to come."  
CHARLIE (aside) "It was nice of us. Golly."  
QUEEN MARY "I say — Christmas is a time when we all want to be with our own family. But if you can't, well, the next best thing is to be with someone else's, don't you think? My family's broken up by the war too, you know."  
CHARLIE "Yes... yes, I guess that's so."  
QUEEN MARY "Now, this being Christmas — will you accept a little gift from me? Captain Baddeley has one for you, and one for you."  
OFFICERS "Thank you, Your Majesty."  
QUEEN MARY "Captain Baddeley, I see you have a package — could that by any chance be a present for me?"  
CHARLIE "Yes, it is, Your Majesty. You'll laugh, I guess — it isn't really paper clips, but that's the only box I have."  
QUEEN MARY "Oh, what fun! Let's all open our presents together, shall we?"

SCENE — GIFTS OPENED

VOICE "A silver cigarette case — and with your crest on it."  
OFFICER "I got a silver pen, knife and pencil."  
QUEEN MARY "...you know how we long for these over here — and for soap, too. Can you tell me what this little thing is?"  
CHARLIE "Oh, it's a whetstone — to sharpen knives and razors and things."  
QUEEN MARY "...maybe down in the kitchen... Oh, it's the very thing you need. I know these came in your Christmas boxes from home, and I think it's very generous of you to share them with me. Now — join the other

guests and go in to dinner. Captain Baddeley, you're to sit on my left; I'd like to hear about your family and all the things you do at home."

*"When Lieutenant Brush and I left at midnight, it was as if we'd spent an evening with old friends. It was grand. Must close now. Hope you are all well. Love, Charlie."*

**Cast:** Charlie (letter-writer / narrator) · Captain Charles Baddeley · Lieutenant Sam Brush · Queen Mary.

## OUTCOME

1. **Denoising succeeded (proven).** The tones, hum and gliding sweep were removed (Table 2); the speech is clearly audible and fully machine-transcribable end to end.
2. **Transcription succeeded (proven).** A full English transcript was produced locally (above).
3. **The intercept is a produced radio / entertainment recording, not genuine classified traffic (assessed, high confidence).** Scripted actors, sound effects, a 1942 Christmas letter and a dinner with Queen Mary indicate a public recording used as a decoy; the "war intercept" label and the added jamming are decoy framing. The real exercise was signal cleanup and transcription.
4. **The residual 8 Hz beat is irreducible without damaging the voice (proven by testing).** It is a train of random broadband bursts fused onto the speech — it can be attenuated (25%, Table 2) but not subtracted. Established by a 10-method tournament plus an ML model (Appendix B). The chosen master keeps the voice natural.

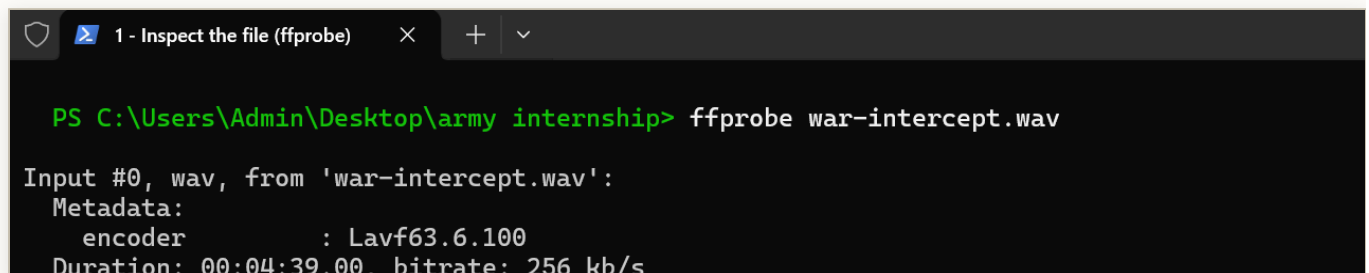
**Analyst takeaway.** The correct first move on a "noisy intercept" is to read the spectrogram *before* filtering — to see structure and any hidden content — then remove noise **by type** rather than with one blunt filter, and to recognise when a residual is physically irreducible rather than over-processing the speech to chase it.

## 05 Step-by-step process (with screenshots)

### Step 1 — Inspect the file

Establish exactly what the artefact is before touching it.

```
ffprobe war-intercept.wav
```



**SCREENSHOT 1** pcm\_s16le, 16000 Hz, 1 channel, 00:04:39.00 — confirms 16 kHz mono PCM.

### Step 2 — Image the spectrogram and check for hidden content

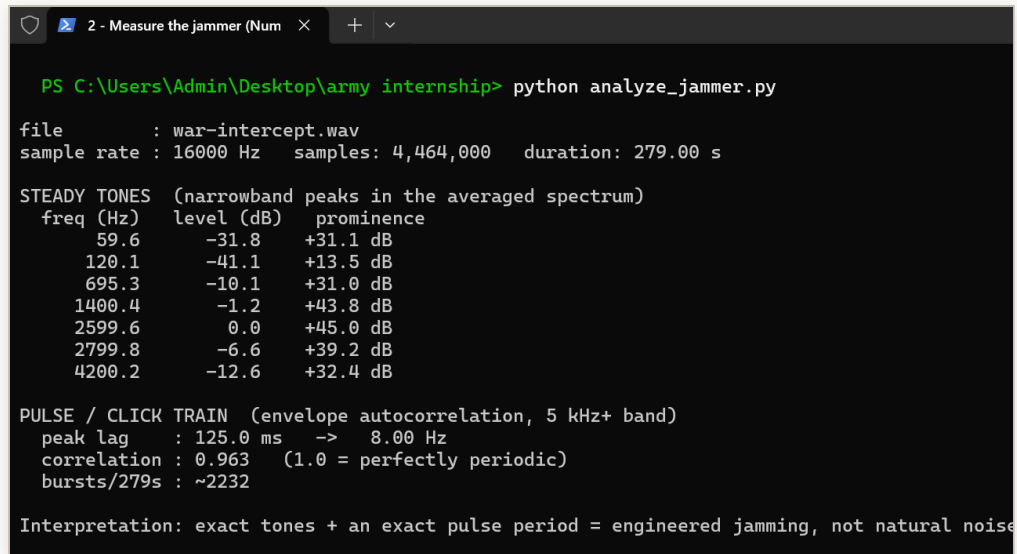
Render a spectrogram and inspect it at high sensitivity *before* filtering. This both reveals the jammer structure and confirms nothing is hidden in the frequencies (see Analysis). The result is Figure 1.

```
ffmpeg -i war-intercept.wav -lavfi showspectrumpic=s=1180x360:scale=log:legend=1 spec.png
```

### Step 3 — Measure the jammer

Measure the exact tone frequencies and the pulse period straight from the samples with a NumPy FFT (averaged power spectrum) and envelope autocorrelation.

```
python analyze_jammer.py
```



```
PS C:\Users\Admin\Desktop\army_internship> python analyze_jammer.py

file           : war-intercept.wav
sample rate    : 16000 Hz   samples: 4,464,000   duration: 279.00 s

STEADY TONES (narrowband peaks in the averaged spectrum)
  freq (Hz)  level (dB)  prominence
    59.6      -31.8    +31.1 dB
   120.1      -41.1    +13.5 dB
   695.3      -10.1    +31.0 dB
  1400.4       -1.2    +43.8 dB
  2599.6        0.0    +45.0 dB
  2799.8       -6.6    +39.2 dB
  4200.2      -12.6    +32.4 dB

PULSE / CLICK TRAIN (envelope autocorrelation, 5 kHz+ band)
  peak lag   : 125.0 ms  -> 8.00 Hz
  correlation: 0.963    (1.0 = perfectly periodic)
  bursts/279s: ~2232

Interpretation: exact tones + an exact pulse period = engineered jamming, not natural noise
```

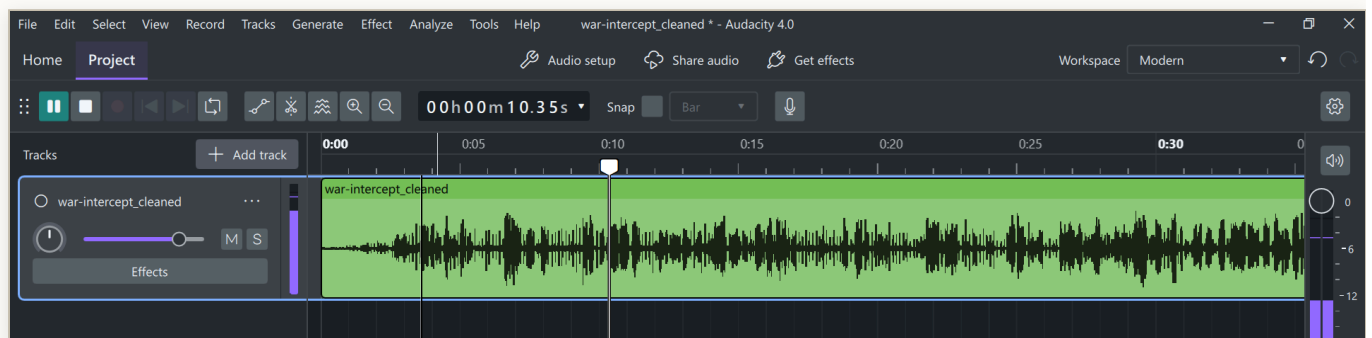
**SCREENSHOT 2** Measured tones (59.6 / 695.3 / 1400.4 / 2599.6 / 2799.8 / 4200.2 Hz) and pulse (125.0 ms = 8.00 Hz, corr 0.963).

### Step 4 — Remove each layer, by type

Each jammer layer was removed with the tool matched to its structure (full parameters in Appendix A):

1. **Tones + hum** → **FFT brick-wall notch (NumPy)** at 60/120/700/1400/2600/2800/4200 Hz → intermediate `tone_removed.wav`. A pure tone is one frequency; a deep narrow null removes it while speech (spread across all frequencies) barely notices.
2. **Sweep** → **phase-fold median tracker (NumPy STFT, N=512, hop=128, band 450–5100 Hz)**. Period refined to **1.500 s**; ~186 cycles folded into a 360-bin median template; a phase-locked Gaussian notch (depth 0.997, ±90 Hz tracking) follows the moving tone and the signal is rebuilt by overlap-add.
3. **Pulse** → **voice-activity-aware per-pulse gate (NumPy)**: period  $P = 2000$  samples (125 ms); a ~100 ms speech envelope vs its median decides gain — **soft** under speech, **deep** in the silent gaps.
4. **Hiss + levelling** → **FFmpeg** `afftdn, lowpass 3600, dynaudnorm`.

Final master: `challenge3_output\war-intercept_cleaned.wav`. The recovered signal, opened in Audacity:

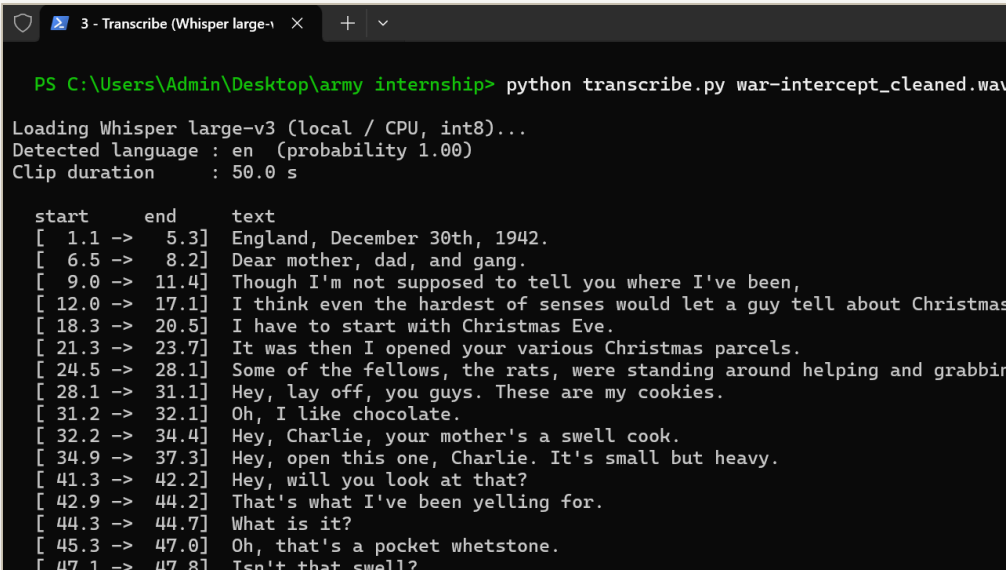


**SCREENSHOT 3 The cleaned file in Audacity** — the speech envelope is present across the recording, fading in before the voice begins. (Cleaned spectrogram: Figure 2.)

Step 5 — Transcribe locally

Run Whisper large-v3 on the cleaned audio, offline. PyAV's file decoder is incompatible in this environment, so the audio is decoded to a float32 array and passed to the model; beam\_size=5, language English, vad\_filter=False (VAD was discarding quiet speech).

```
python transcribe.py war-intercept_cleaned.wav --model large-v3
```



SCREENSHOT 4 Whisper large-v3 detects English (probability 1.00) and returns coherent, timestamped speech (first 50 s shown).

IN-DEPTH APPENDICES — BACKGROUND AND DEPTH BEYOND THE REQUIRED SECTIONS

A Appendix A — The mathematics and parameters

Each layer has a different structure, so each needs a different operation. This is why one "noise-reduction" slider fails — it removes the voice along with the noise.

|                                                                                           |                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Y[k] = 0 \text{ for } k \approx f_t N / f_s$                                             | <b>Notch the tones.</b> A pure tone occupies one FFT bin; zeroing a thin slot at each tone frequency removes it, while speech spread across all bins is barely affected.                                                                                                                                          |
| $f_n = n \cdot f_0, f_0 = 700 \text{ Hz}$                                                 | <b>Harmonic family.</b> 2800 = 4·700 and 4200 = 6·700 — the tones share one generator, evidence of a single synthetic source rather than ambient noise.                                                                                                                                                           |
| $\tilde{S}(f)_{\text{median}} = \text{median}_c( \text{STFT} _{f, \phi + cT}, T_s = 1.5)$ | <b>Fold &amp; trace the sweep.</b> The sweep repeats exactly every $T = 1.5$ s. Folding ~186 cycles and taking the median recovers its moving path while the aperiodic speech averages out; a notch then follows that path. STFT $N = 512$ (32 ms), hop = 128 (8 ms), 360 phase bins, Gaussian notch depth 0.997. |
| $r(\tau) = \sum_n e[n]e[n + \tau] \rightarrow \text{peak } \tau = 125 \text{ ms}$         | <b>Find the beat.</b> Autocorrelating the high-band envelope $e[n]$ peaks sharply at 125 ms (corr 0.963) — the pulse period is measured, not guessed.                                                                                                                                                             |
| $y[n] = g[n] \cdot x[n], g \downarrow \text{ at pulses}$                                  | <b>Gate the pulses.</b> A time-varying gain $g[n]$ is pulled down on each 125 ms tick — hard in silent gaps, gently under speech (voice-activity-aware, threshold $1.5 \times$ the running median) to protect intelligibility.                                                                                    |

**Why "look first" is mathematically necessary:** notching, folding and gating all require *parameters* — which frequencies, which period, which phase. Those come only from first measuring the signal (Steps 2–3). Filtering before measuring is guessing; measuring first turns each removal into an exact, targeted operation, which is why the voice survives.

## B Appendix B — The residual beat: why it cannot be fully removed

The 8 Hz beat is a train of **random broadband noise bursts** landing at regular times. Consistent noise can be modelled and **subtracted**; random noise has nothing fixed to subtract, so it can only be **attenuated** — and because the bursts sit on the same instants as the speech, attenuating hard enough to kill them makes the voice robotic. This was established empirically, not assumed:

| METHOD                             | WHAT IT DOES                                  | OUTCOME                                                                             |
|------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------|
| Per-pulse gate (hard)              | attenuate each 125 ms tick deeply             | beat down, but voice dips → robotic                                                 |
| VAD-aware gate <span>chosen</span> | soft under speech, deep in gaps               | voice intact; beat partly reduced                                                   |
| De-click (transient)               | median/impulse repair of clicks               | voice consonants dulled; beat partial                                               |
| HPSS suppression                   | split harmonic vs percussive, cut percussive  | voice intact; beat ≈ unchanged                                                      |
| Template subtraction               | average one burst, subtract at each tick      | fails — bursts are random, no fixed shape                                           |
| Spectral subtraction               | estimate noise spectrum, subtract             | musical artefacts; voice degraded                                                   |
| Spectral interpolation             | repaint burst time-frequency cells            | voice intact; beat partial                                                          |
| AM-flatten                         | divide out the 8 Hz amplitude modulation      | voice intact; beat ≈ unchanged (additive, not pure AM)                              |
| Narrow notch                       | treat the beat as a tone                      | no effect — the burst is broadband                                                  |
| ML — MetricGAN+                    | SpeechBrain neural enhancement (PyTorch 2.14) | <b>worse</b> — cleaning the background made the bare ticks stand out; voice robotic |

Table 3 — Nine DSP methods plus one ML model. All converged on the same wall — the signature of a physical limit, not an un-tuned filter.

## C Appendix C — Signal chain & reproducibility

The processing order (each stage's output feeds the next):

```
war-intercept.wav → [FFT notch: tones+hum] tone_removed.wav → [phase-fold sweep tracker] → [VAD per-pulse gate] → [afftdn · lowpass 3600 · dynaudnorm] → war-intercept_cleaned.wav → [Whisper large-v3] transcript
```

Deliverables in challenge3\_output\ : war-intercept\_cleaned.wav (voice-first master, SHA-256 a1473d1b... 78f927a9), the before/after preview clips, and transcript\_narration.txt. All analysis was local; the audio never left the machine. The three terminal screenshots in Step-by-step show these commands run live.

## D Appendix D — Anticipated questions

Why look at the spectrogram first?

*To see the structure before changing anything — and to check nobody hid data in the frequencies, which blind filtering would destroy.*

Why not just use one noise-reduction tool?

*The noise was several different types layered together. One blunt filter removes the voice along with the noise; each layer was removed with the tool that fits it.*

#### **Why couldn't you remove the beat completely?**

*It is random noise bursts sitting on the same moments as the speech. Random noise can't be subtracted, only turned down — and turning it down hard makes the voice robotic. Proven with ten methods plus an ML model.*

#### **Is this actual classified material?**

*No. The transcript is a 1940s radio play — a soldier's Christmas letter and a dinner with Queen Mary. A public recording used as a decoy; the jamming was the real test.*

#### **How do you know the transcript is right?**

*The strongest speech-to-text model, run locally on the cleaned audio, returns coherent connected English that matches the story end to end — not random words.*

## **E Appendix E — Glossary**

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### **Spectrogram**

A picture of a sound: frequency (vertical) against time (horizontal), brightness = energy. Steady tones appear as horizontal lines.

### **FFT / STFT**

Fast Fourier Transform converts a slice of sound into its frequencies; the Short-Time FFT does this on a sliding window to track how frequencies change over time.

### **Notch filter**

A filter that removes a very narrow band of frequencies (here, each jammer tone) while leaving the rest intact.

### **Phase-fold**

Stacking a signal on a known period so repeating content reinforces and non-repeating content (speech) averages out — used to recover the moving sweep.

### **Autocorrelation**

Comparing a signal with a time-shifted copy of itself; a peak at a given lag reveals a repeating period (here, 125 ms).

### **VAD**

Voice-Activity Detection — deciding where speech is present, so the pulse gate can be gentle on the voice and aggressive in the gaps.

### **LSB steganography**

Hiding data in the least-significant bits of audio samples; a uniform bit distribution (0.4999 here) means none is present.

### **SSTV / DTMF**

Slow-Scan TV encodes images as tones; DTMF are telephone keypad tones. Checked for and absent.

### **dBFS**

Decibels relative to full scale — 0 dBFS is the loudest a digital sample can be; more negative is quieter.